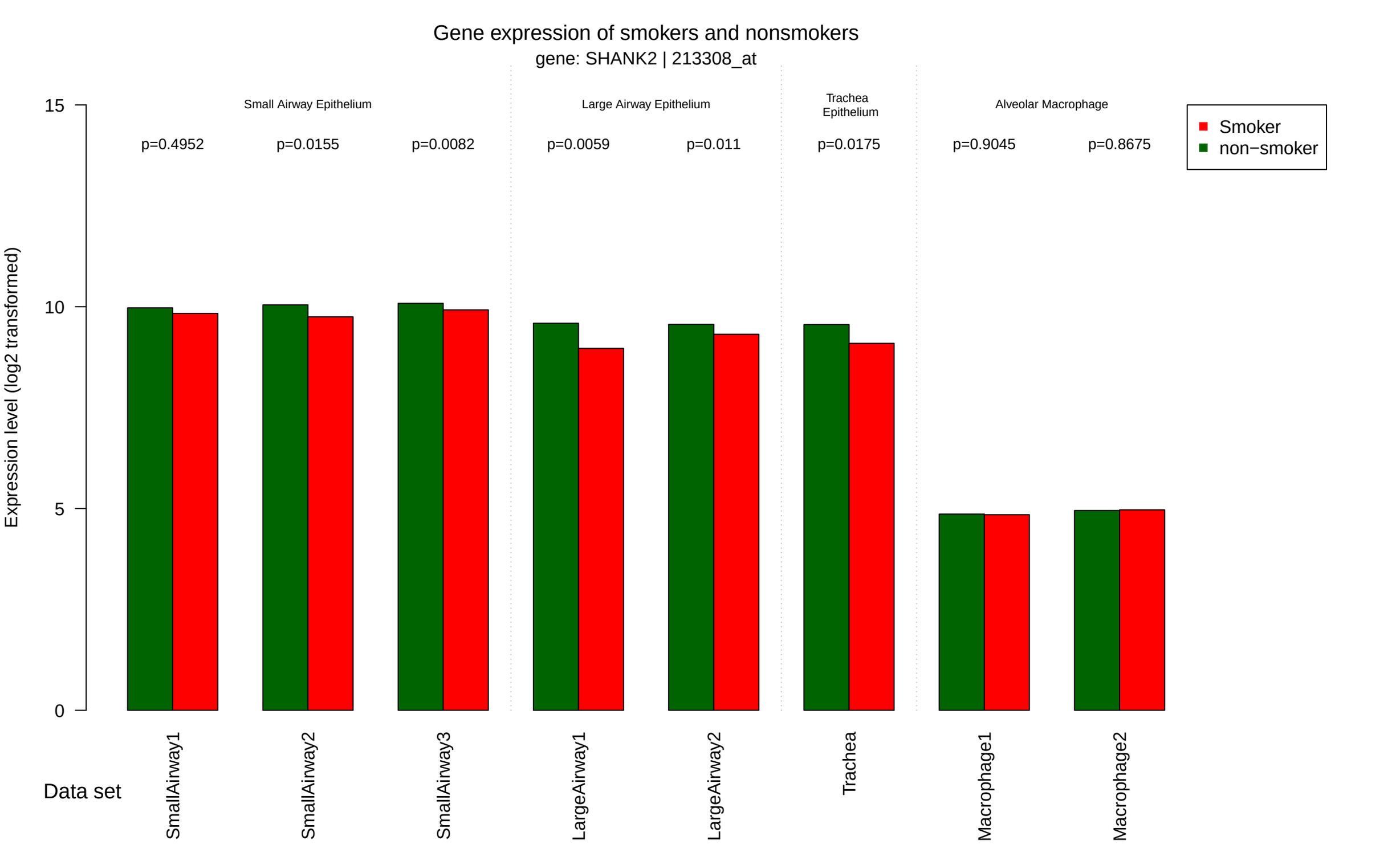
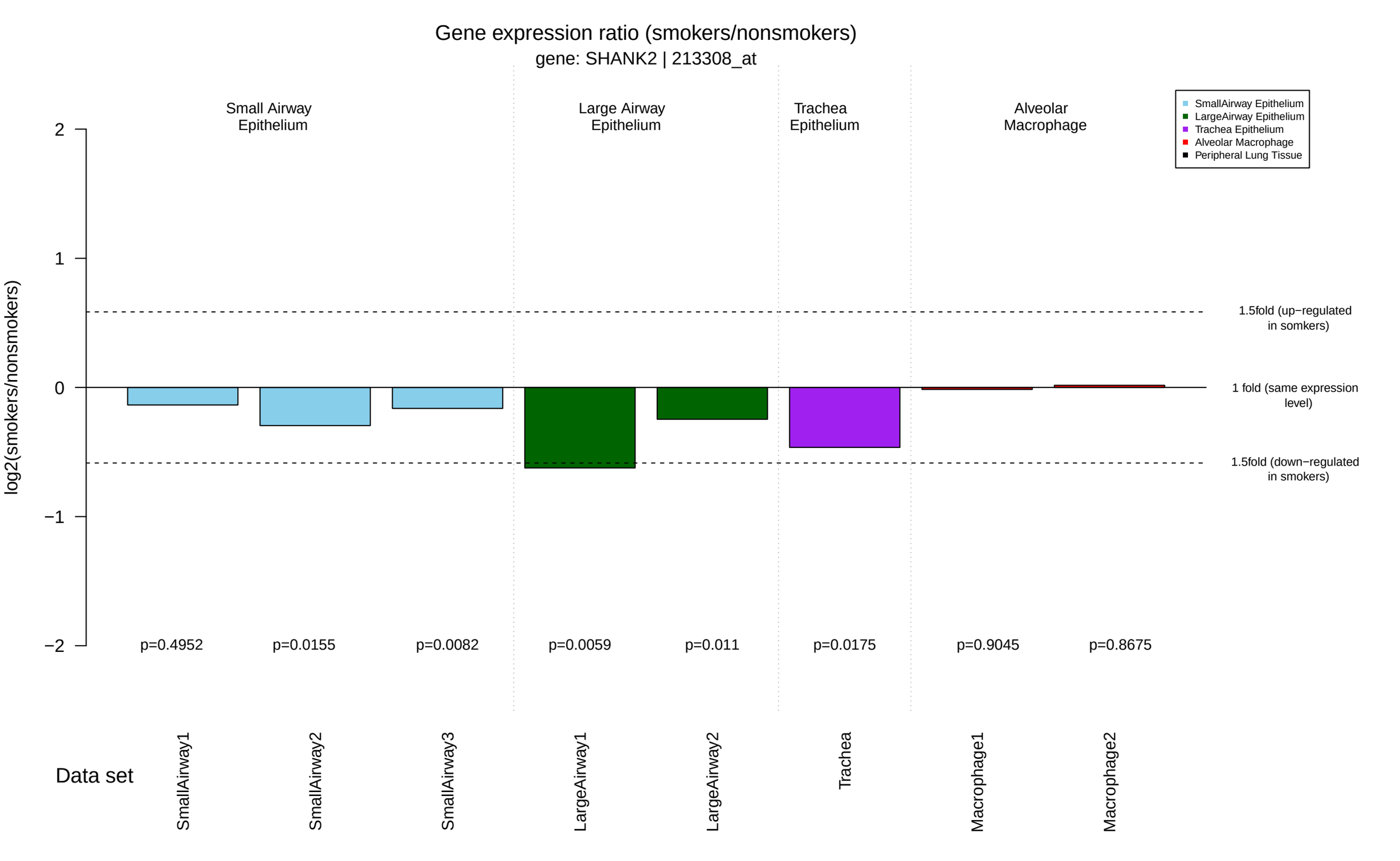
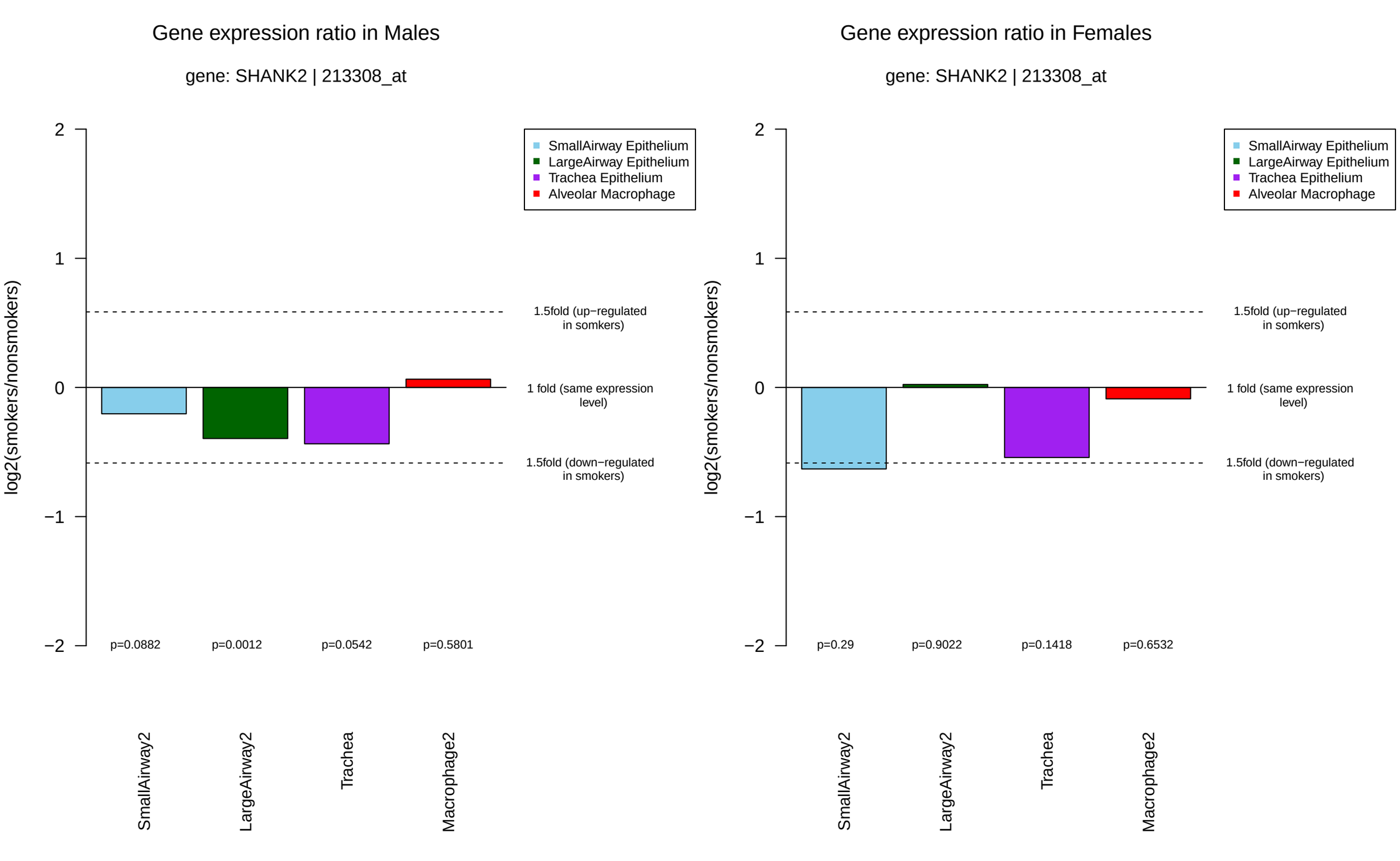
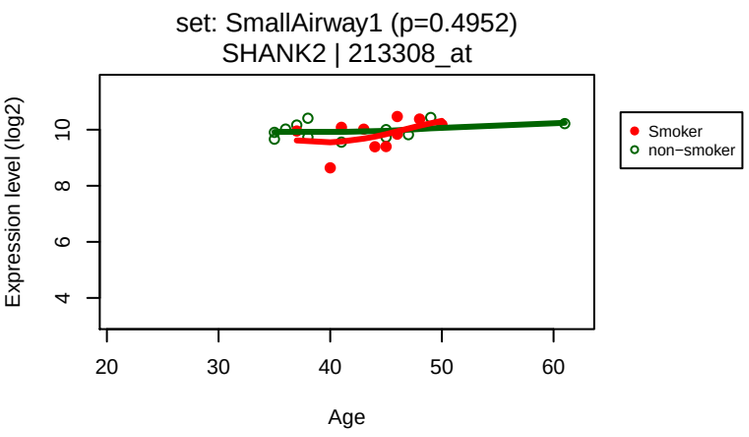
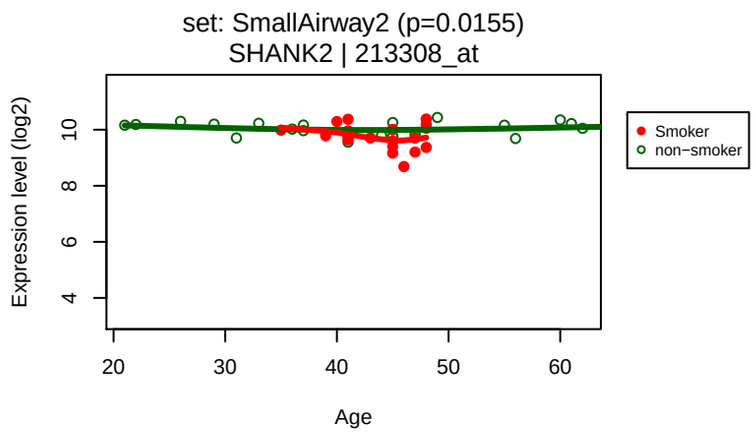
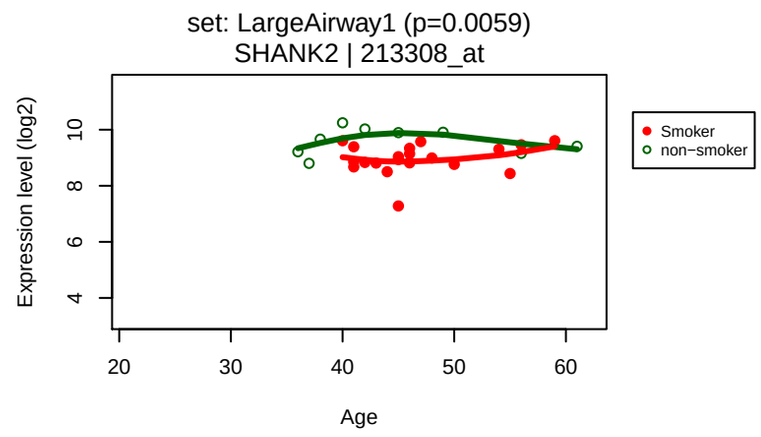

Gene expression of smokers and nonsmokers
gene: SHANK2 | 213308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4952
p=0.0155
p=0.0082
p=0.0059
p=0.011
p=0.0175
p=0.9045
p=0.8675
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

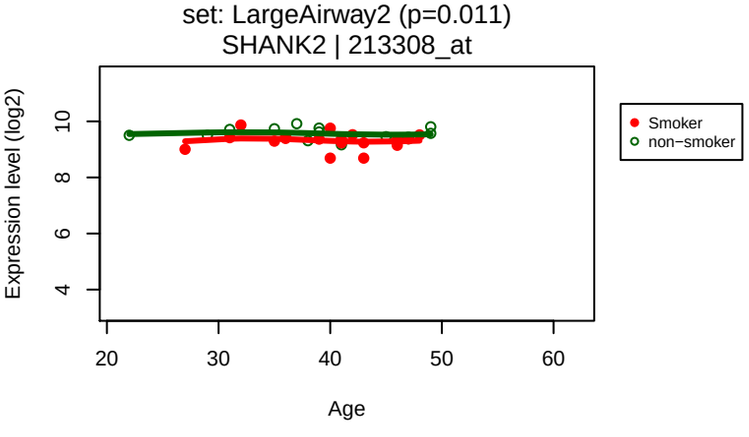

Gene expression ratio (smokers/nonsmokers)
gene: SHANK2 | 213308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4952
p=0.0155
p=0.0082
p=0.0059
p=0.011
p=0.0175
p=0.9045
p=0.8675
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

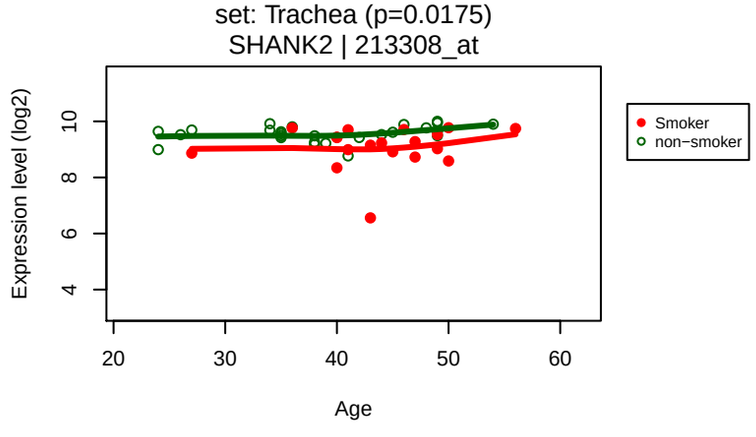

Gene expression ratio in Males
gene: SHANK2 | 213308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0882
p=0.0012
p=0.0542
p=0.5801
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SHANK2 | 213308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.29
p=0.9022
p=0.1418
p=0.6532
SmallAirway2
LargeAirway2
Trachea
Macrophage2

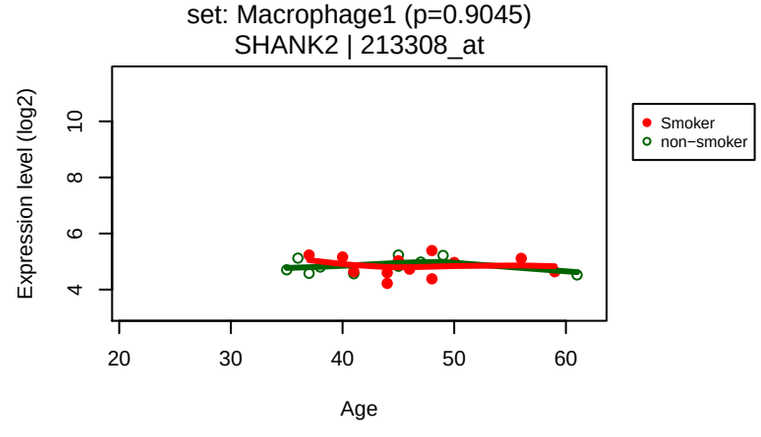

set: SmallAirway1 (p=0.4952)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker

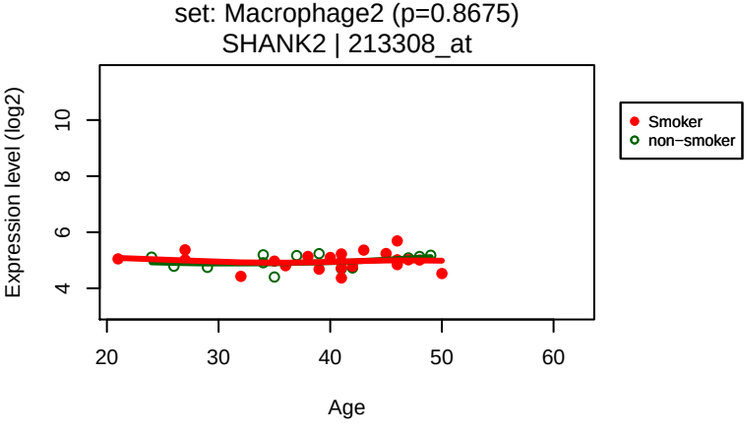

set: SmallAirway2 (p=0.0155)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker

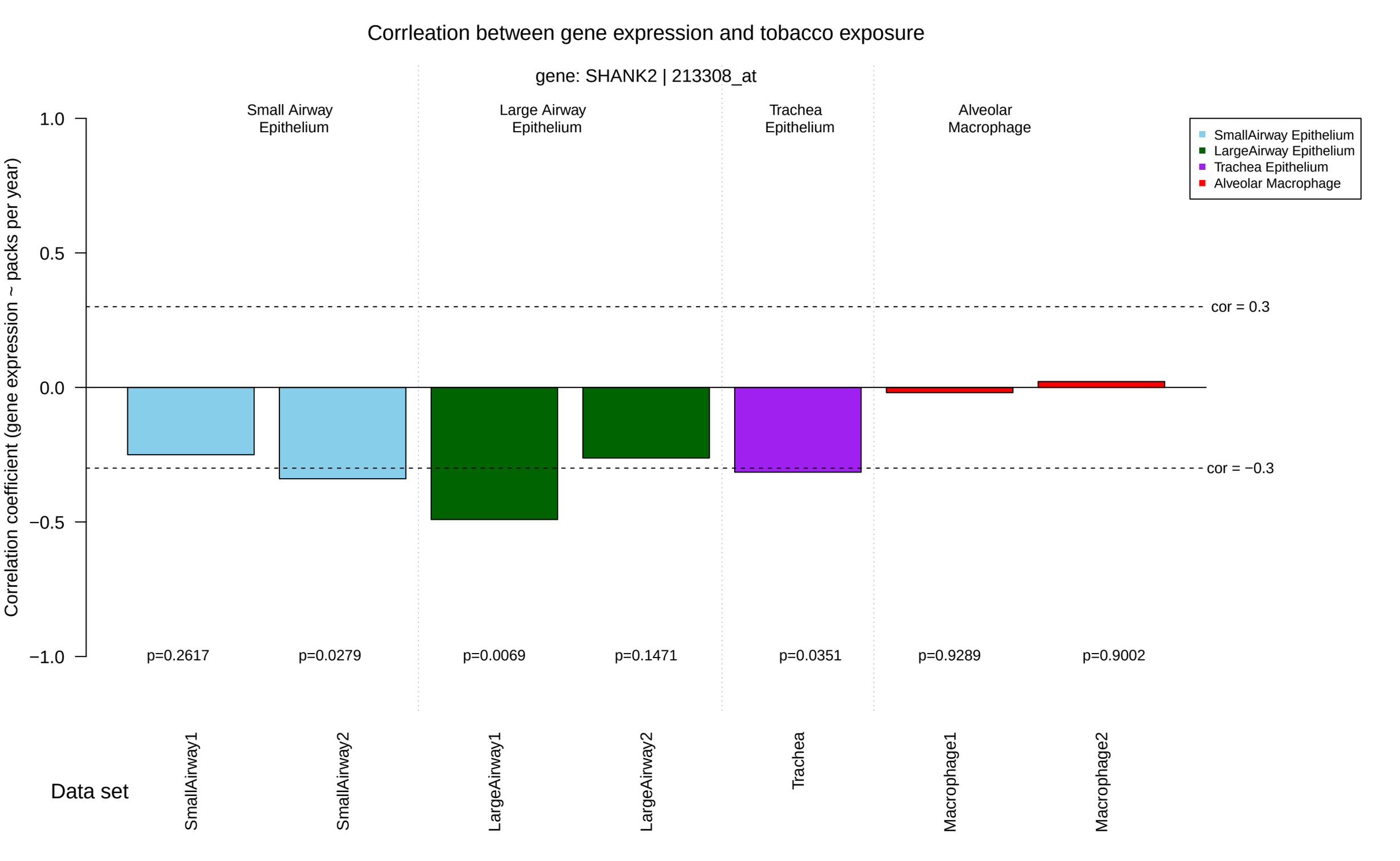

set: LargeAirway1 (p=0.0059)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.011)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0175)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9045)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8675)
SHANK2 | 213308_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SHANK2 | 213308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2617
p=0.0279
p=0.0069
p=0.1471
p=0.0351
p=0.9289
p=0.9002
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2